

Task 1: File copy using open(), read(), write(), lseek() and close()

Code:

```
GNU nano 7.2 copy_lowlevel.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <errno.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    int src = open(argv[1], O_RDONLY);
    if (src == -1) {
        perror("Source open");
        return 1;
    }

    int dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest == -1) {
        perror("Destination open");
        close(src);
        return 1;
    }

    char buffer[4096];
    ssize_t bytes;

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
        ssize_t written = 0;
        while (written < bytes) {
            ssize_t n = write(dest, buffer + written, bytes - written);
            if (n == -1) {
                perror("Write");
                close(src);
                close(dest);
                return 1;
            }
            written += n;
        }

        if (bytes == -1)
            perror("Read");

        close(src);
        close(dest);

        if (bytes == -1)
            return 1;

        printf("File copied successfully.\n");
        return 0;
    }
}
```

Output:

```
karthik@Karthikeya: ~/ossp/v  x  +  v
karthik@Karthikeya:~$ mkdir -p ~/ossp/week9
karthik@Karthikeya:~$ cd ~/ossp/week9
karthik@Karthikeya:~/ossp/week9$ nano copy_lowlevel.c
karthik@Karthikeya:~/ossp/week9$ gcc copy_lowlevel.c -o copy_lowlevel
karthik@Karthikeya:~/ossp/week9$ echo "Operating Systems Practical Week 9" > input.txt
./copy_lowlevel input.txt output.txt
cat output.txt
File copied successfully.
Operating Systems Practical Week 9
karthik@Karthikeya:~/ossp/week9$ nano seek_demo.c
karthik@Karthikeya:~/ossp/week9$ gcc seek_demo.c -o seek_demo
karthik@Karthikeya:~/ossp/week9$ ./seek_demo
Data after seeking: stems Practical Week
karthik@Karthikeya:~/ossp/week9$
```

Task 2: Compare system calls with standard library file I/O

Code:

```
karthik@Karthikeya: ~/ossp/v  +  GNU nano 7.2  compare_io.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <time.h>

#define SIZE (1024 * 1024)

int main() {
    char buffer[4096];
    struct timespec start, end;

    int fd = open("input.txt", O_RDONLY);
    if (fd == -1) {
        perror("open");
        return 1;
    }

    clock_gettime(CLOCK_MONOTONIC, &start);

    while (read(fd, buffer, sizeof(buffer)) > 0);

    clock_gettime(CLOCK_MONOTONIC, &end);
    printf("System call I/O time: %.6f seconds\n",
        (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9);

    close(fd);

    FILE *fp = fopen("input.txt", "rb");
    if (!fp) {
        perror("fopen");
        return 1;
    }

    clock_gettime(CLOCK_MONOTONIC, &start);

    while (fread(buffer, 1, sizeof(buffer), fp) > 0);

    clock_gettime(CLOCK_MONOTONIC, &end);
    printf("Standard library I/O time: %.6f seconds\n",
        (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9);

    fclose(fp);
    return 0;
}
```

Output:

```
karthik@Karthikeya:~/ossp/week9$ nano compare_io.c
karthik@Karthikeya:~/ossp/week9$ gcc compare_io.c -o compare_io
karthik@Karthikeya:~/ossp/week9$ ./compare_io
System call I/O time: 0.000002 seconds
Standard library I/O time: 0.000007 seconds
karthik@Karthikeya:~/ossp/week9$ |
```